

Data Flow Diagram

Date	28 June 2026
Team ID	SWTID-2026-4993
Project Name	OptiCrop: Smart Agricultural Production Optimization Engine 2
Maximum Marks	2 Marks

Data Flow Diagram

The Data Flow Diagram represents the flow of information in the OptiCrop system. User input is collected, preprocessed, analyzed using a machine learning model, and the predicted crop recommendation is displayed to the user. This process ensures accurate crop prediction based on soil and environmental conditions. .

DFD Symbol Legend

Symbol	Name	Description
Oval / Rounded shape	External Entity	A person, organization, or system outside the project that sends or receives data (e.g., Farmer, Researcher).
Rectangle with numbered header	Process	An activity that transforms incoming data into outgoing data (e.g., Validate Input, ML Prediction Engine).
Rectangle (solid fill, no number)	Data Store	A place where data is held for later use (e.g., Agricultural Dataset CSV, ML Model Store .pkl).
Labeled arrow	Data Flow	The movement of data between entities, processes, and data stores, labeled with what data is being passed.

DATA FLOW DIAGRAM (DFD) – OPTICROP



